POLITECNICO
MILANO 1863

EFFECT OF CROSS WIND IN VEHICLE DYNAMIC STABILITY

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1. Introduction

2. Objectives

2. Objectives

Different phases of the study

Wind

- 2 Wind velocities (25 & 30 m/s)
- 3 turbulence intensities each (7, 14 & 25 %)

Truck

- Loading factor variations (λ)
- Roll stiffness distribution (τ)

Maneuvers

- Single & Double lane change
- Right and left turns

3. Methodology

3.Methodology Assumptions and

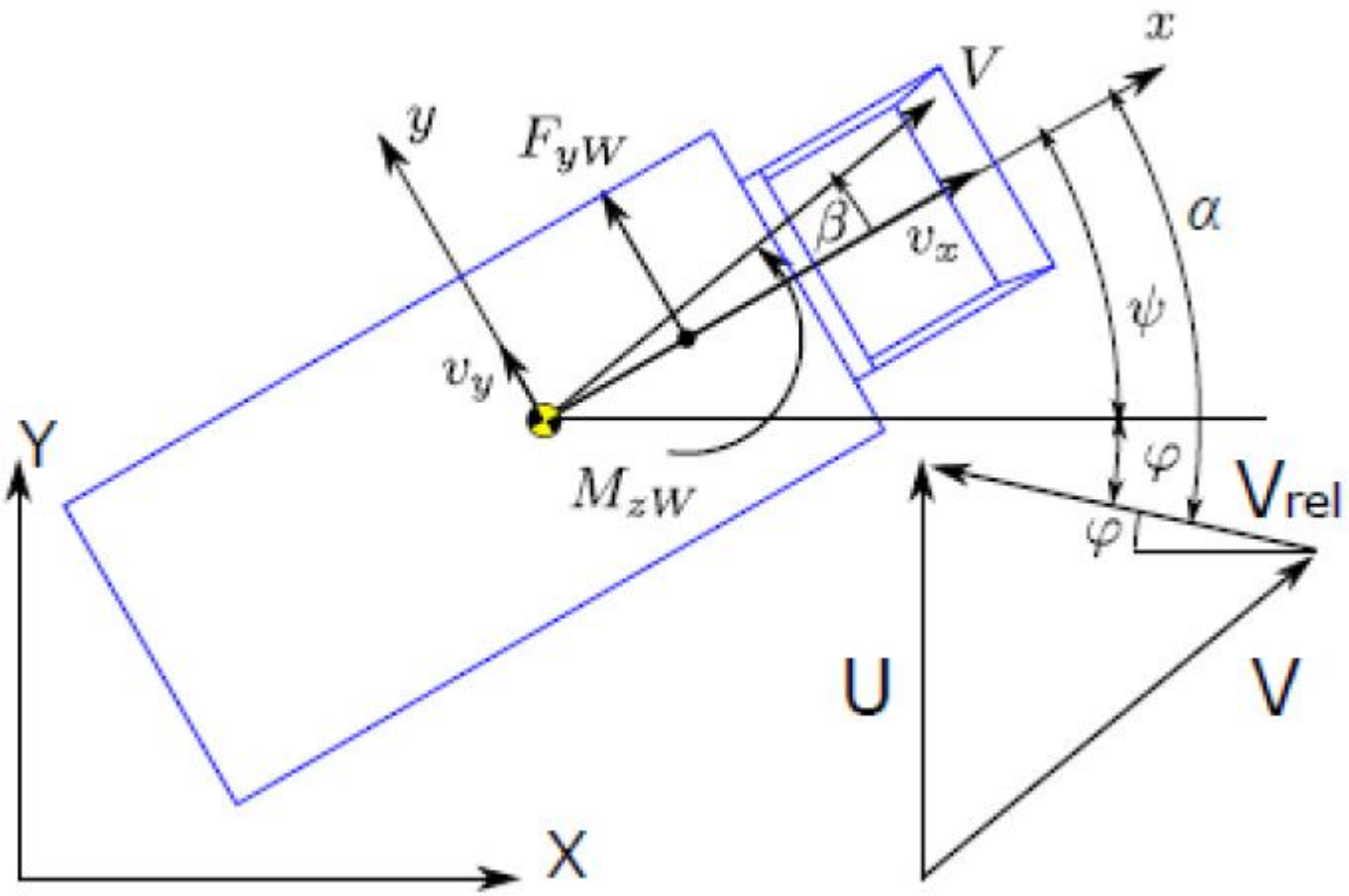
Important considerations

- Wind velocity in Y direction
- Only Mx and Mz considered
- Constant vx velocity
- No vertical displacement

-**Pacejka model** for the lateral tire forces.

-Given **aerodynamic coefficients** function of α.

-**Wind history** along 1000 meters and over 100 seconds.

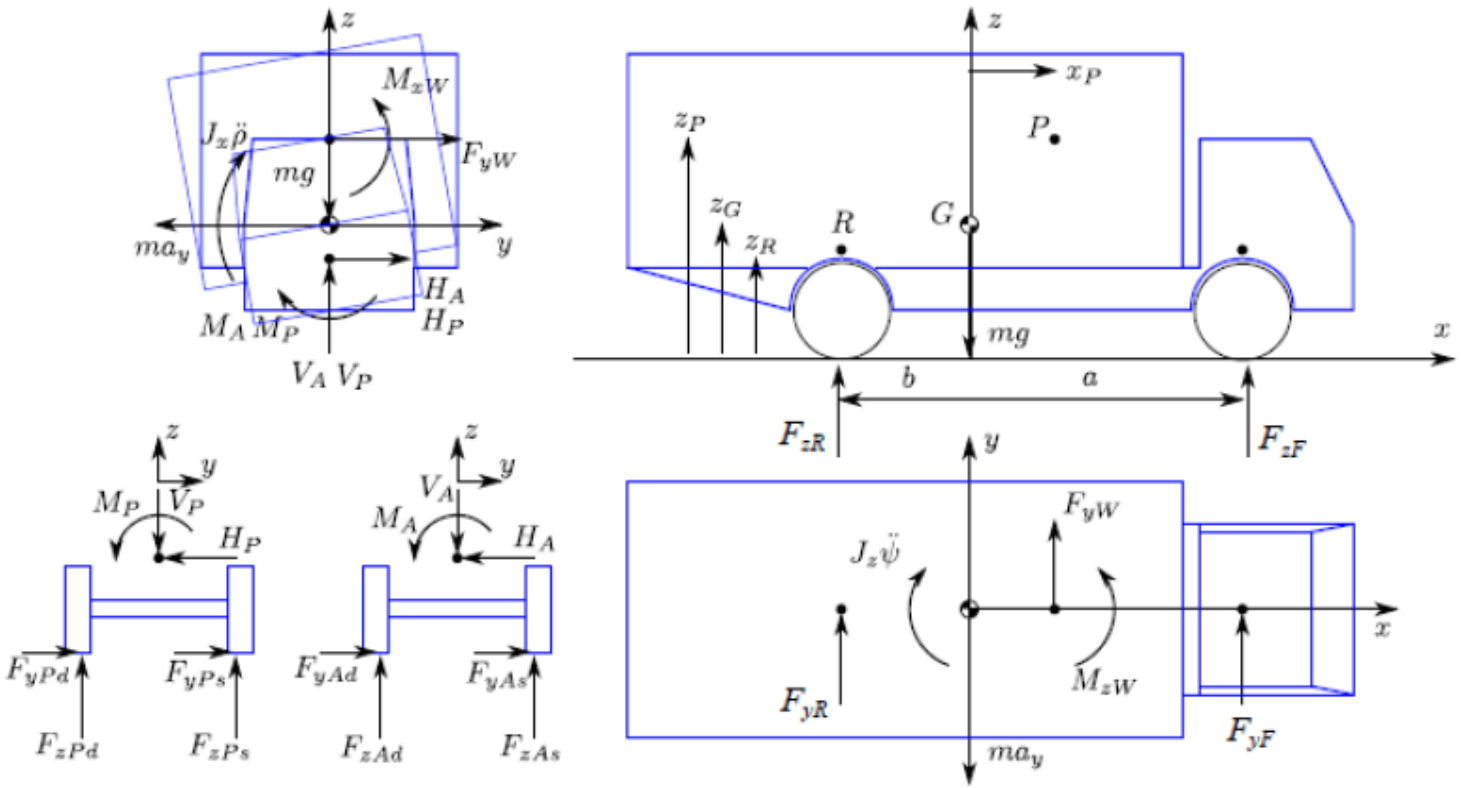


3. Methodology

Different cases

Rollover

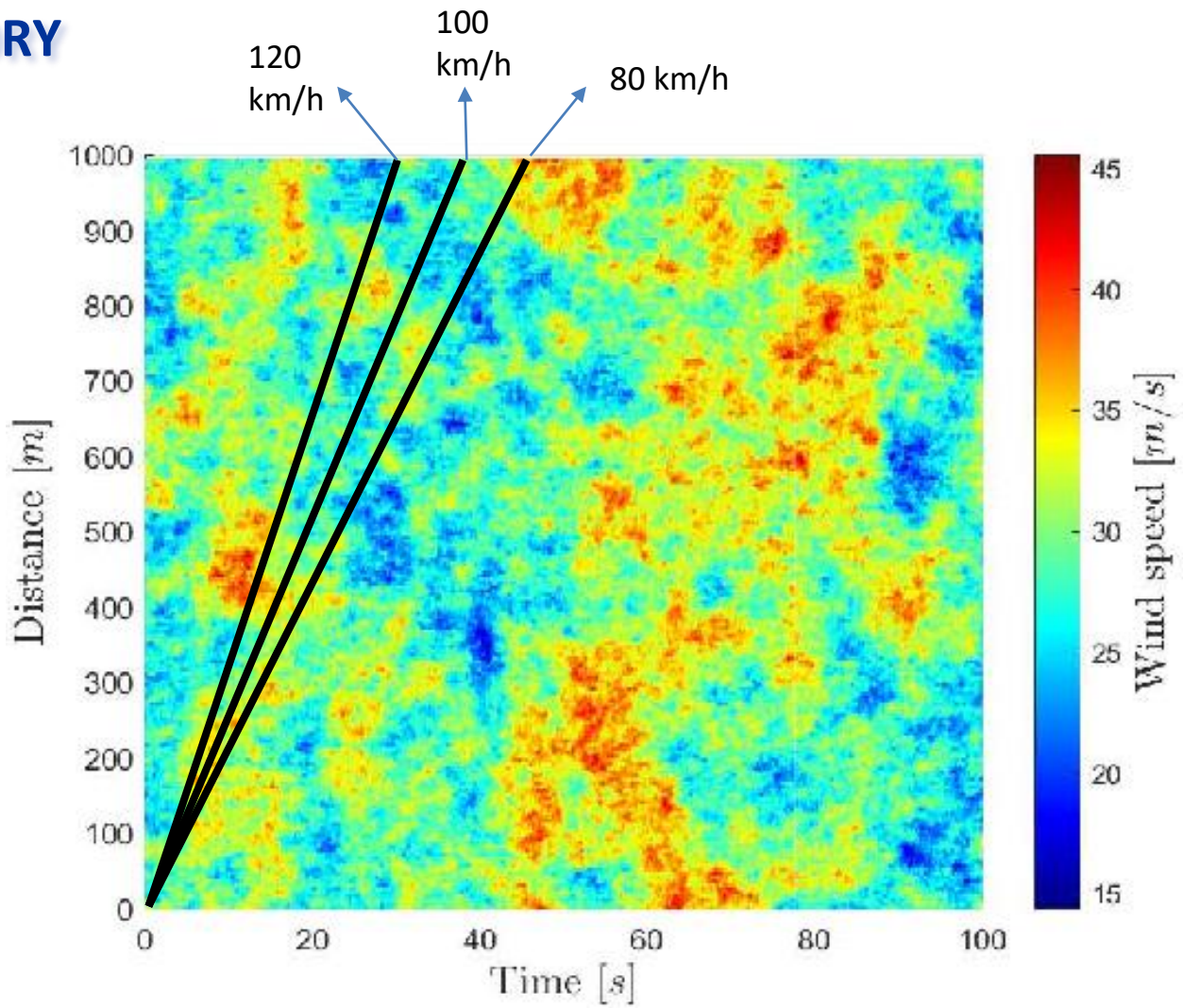
- It is considered to occur when the **reaction force** at any of the tires becomes zero. As a safety margin, a threshold of **10% of the static load** is used instead.



4. Results

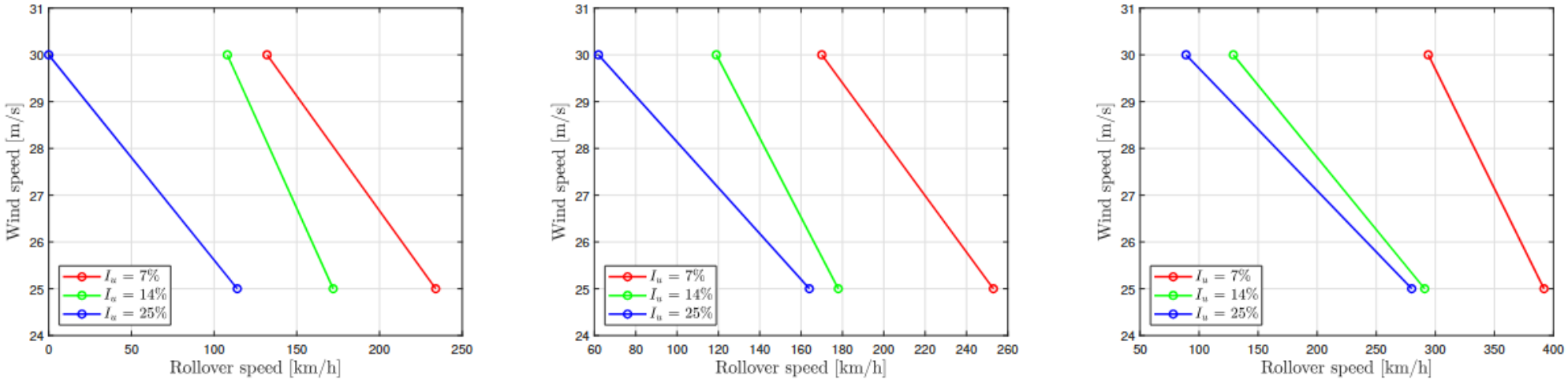
4. Results

Wind history



4. Results

Characteristic wind curve

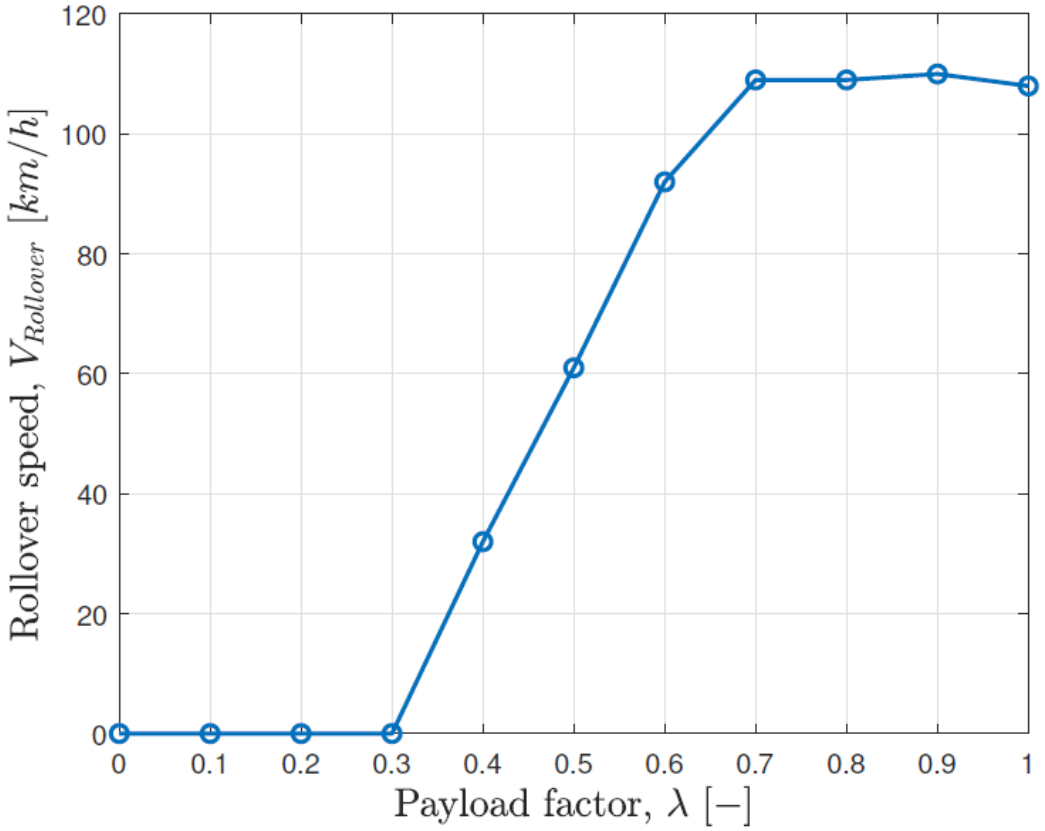


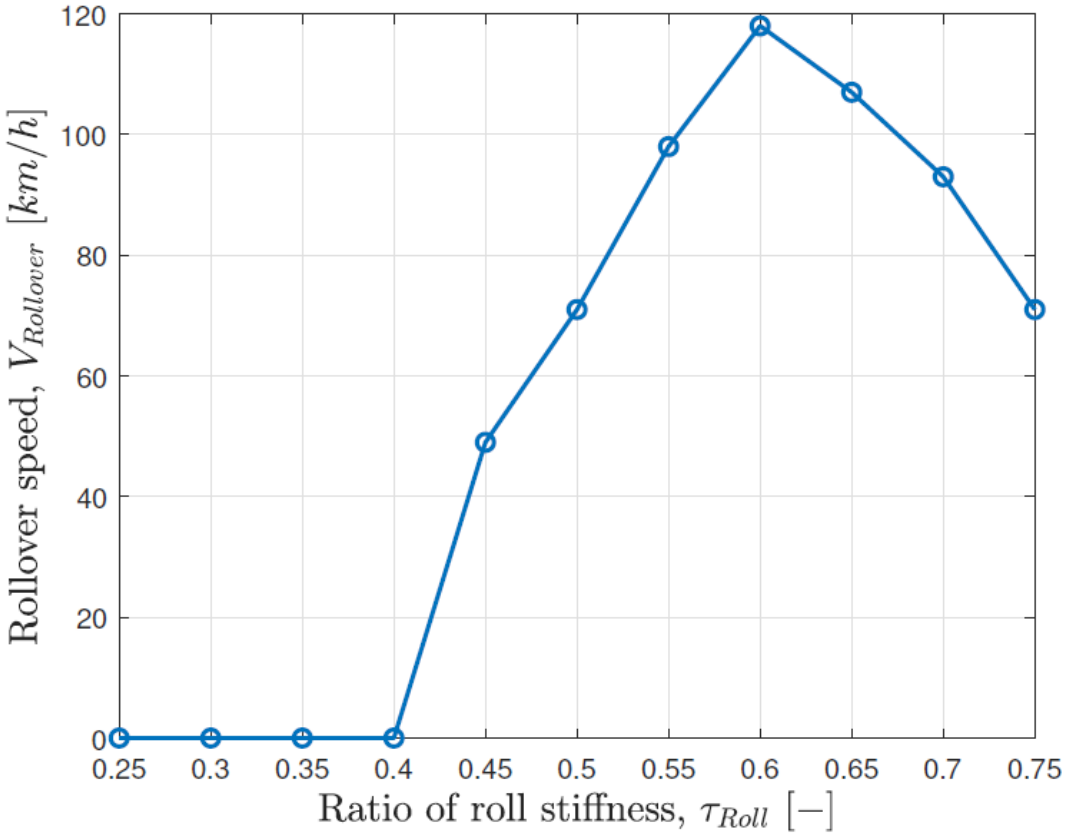
(a) Dry. (b) Wet. (c) Snow.

	Turbulence index								
	7%			14%			25%		
Wind Speed $[m/s]$	Dry	Wet	Snow	Dry	Wet	Snow	Dry	Wet	Snow
25	234	253	392	172	178	291	114	164	280
30	132	170	294	108	119	129	0	62	89

4. Results

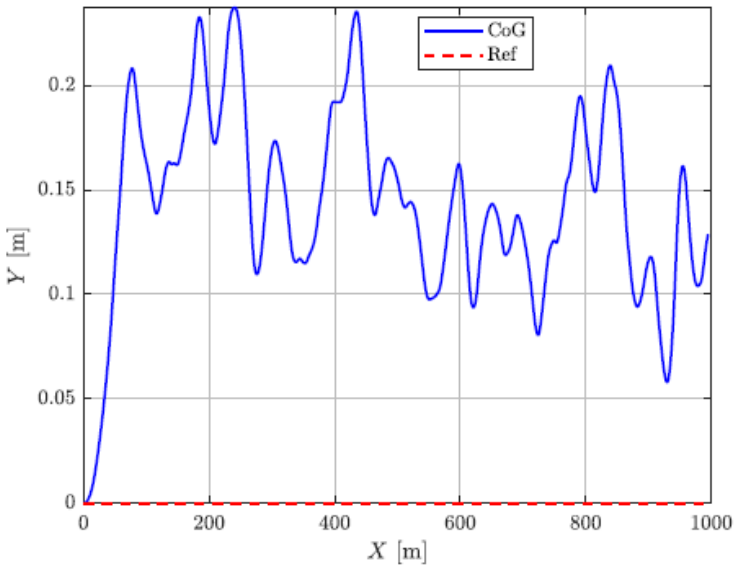
Effect of payload factor (λ) and roll stiffness distribution (τ) on rollover behavior



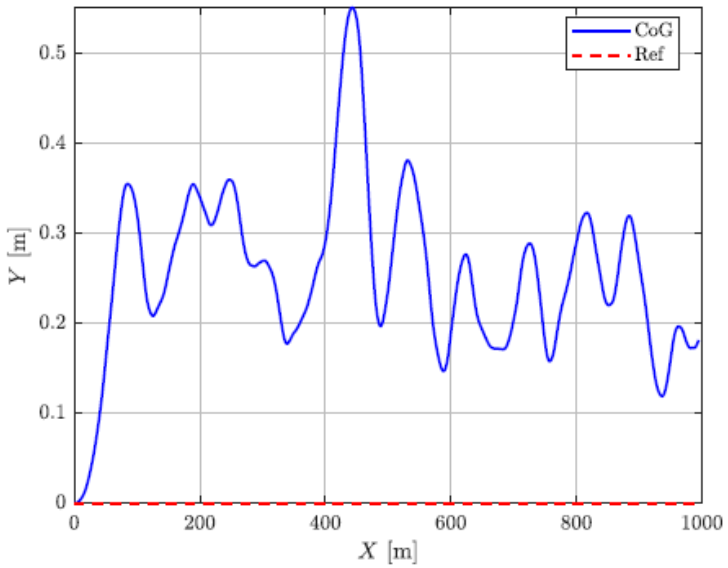


4. Results

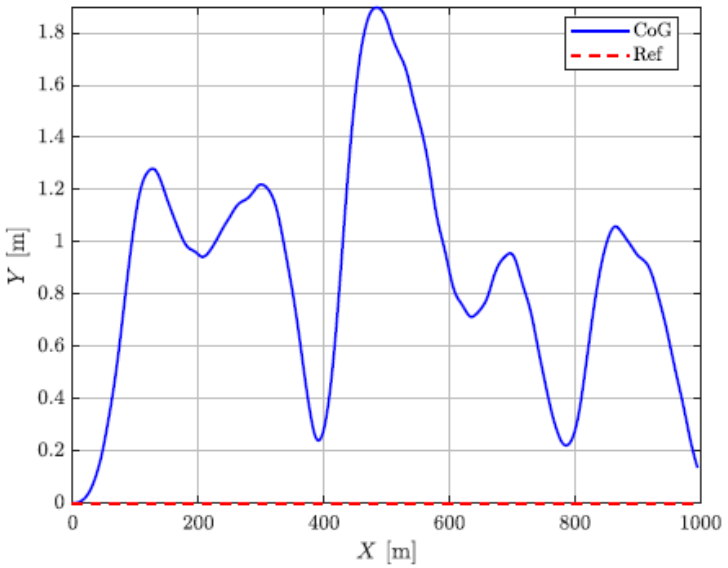
Displacement of the center of gravity in different road conditions



(a) Dry.



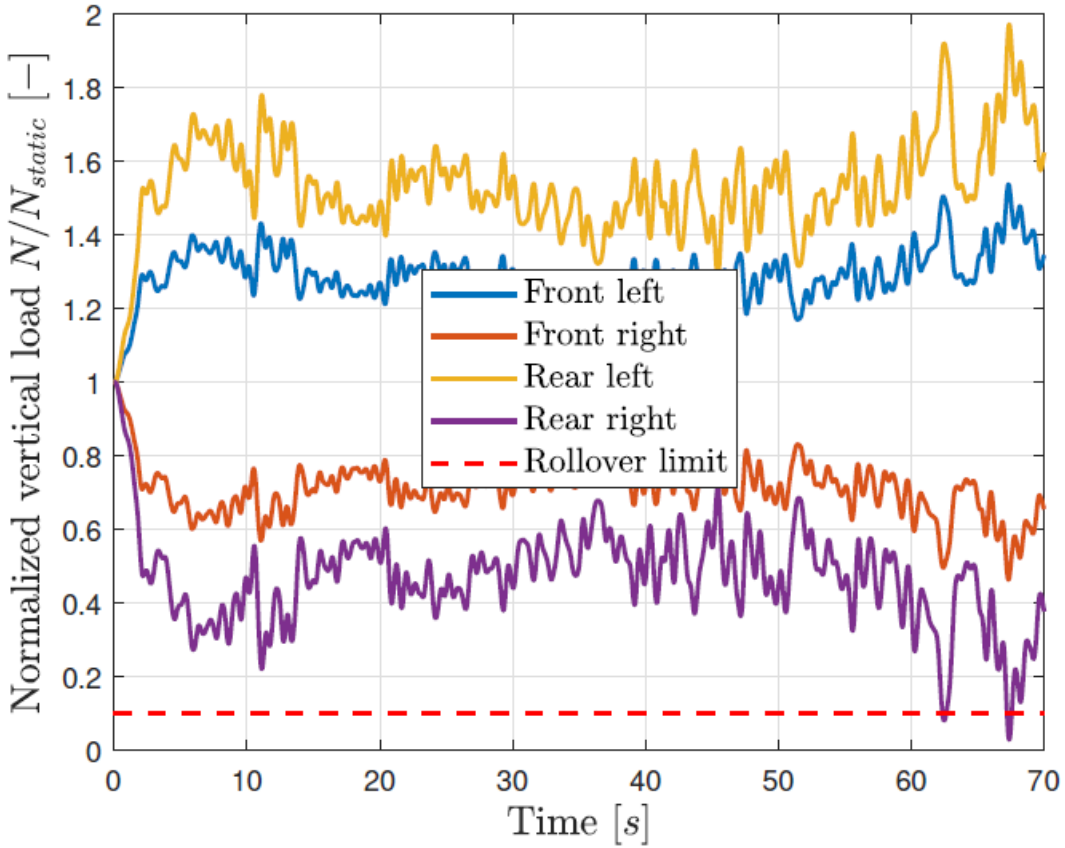
(b) Wet.



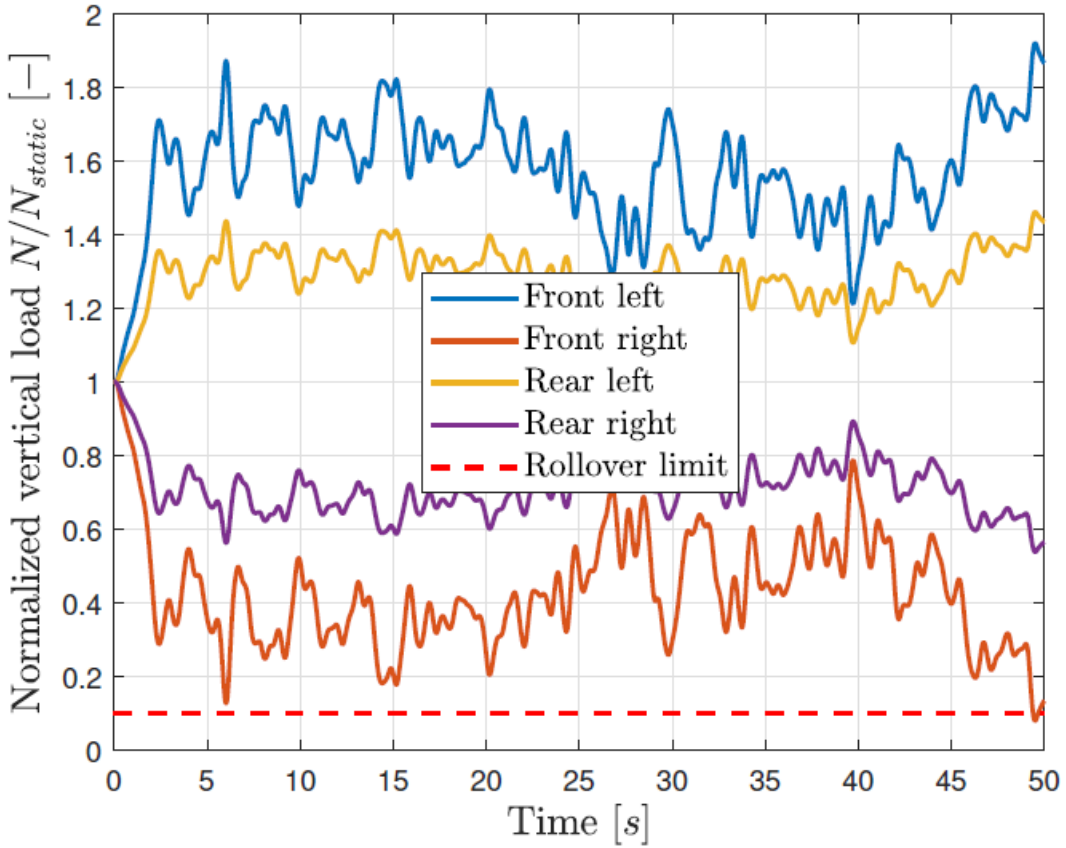
(c) Snow.

4. Results

Load on each wheel with different roll stiffness distribution (τ)



(a) $\tau_{Rollover} = 0.45.$

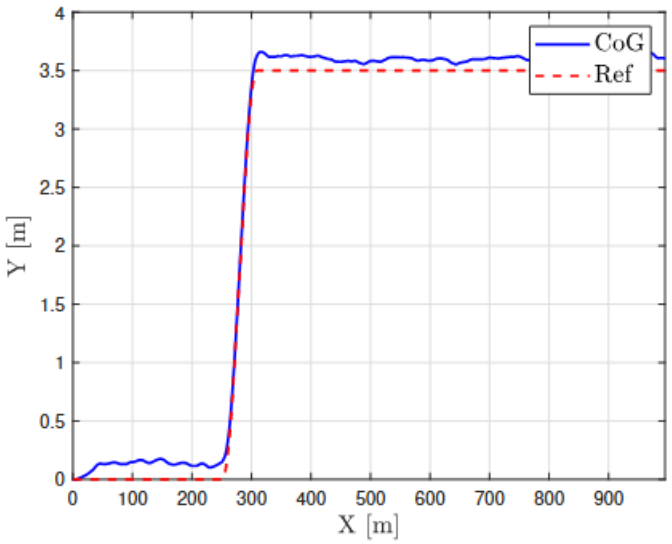


(b) $\tau_{Rollover} = 0.75.$

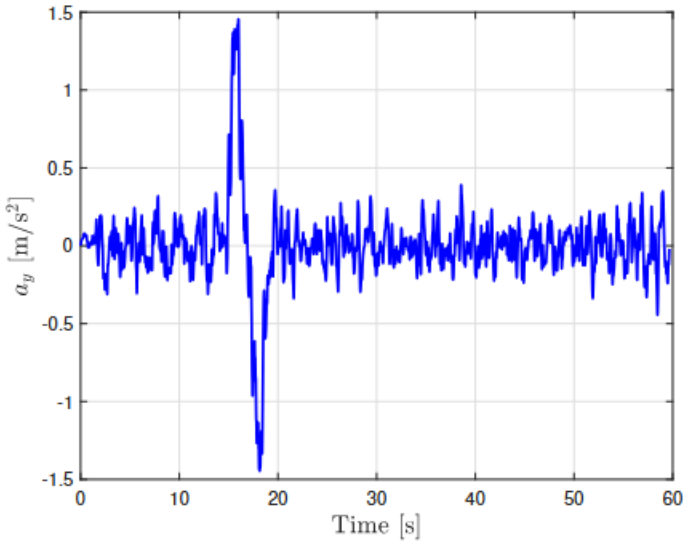
4. Results

Influence of turning maneuvers on rollover speed

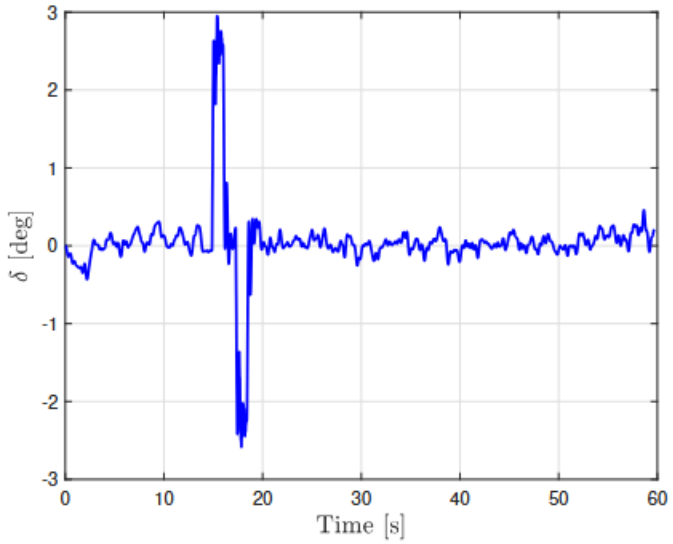
$$\delta = k_p \left(Y_{\text{ref}} - Y_{\text{oss}}\right) + k_d \left(\dot{Y}_{\text{ref}} - \dot{Y}_{\text{oss}}\right)$$



(a) Trajectory.



(b) Lateral acceleration.



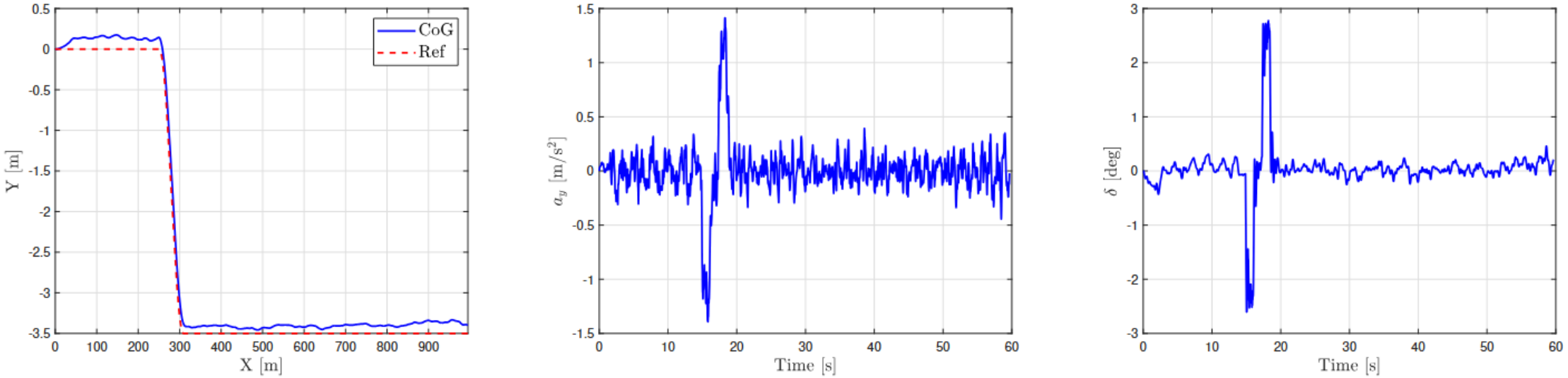
(c) Steering angle.

Left turn - Dry road - Wind ($V = 30m/s$, $I_u = 14\%, L_u = 150m$)

4. Results

Influence of turning maneuvers on rollover speed

$$\delta = k_p \left(Y_{\text{ref}} - Y_{\text{oss}}\right) + k_d \left(\dot{Y}_{\text{ref}} - \dot{Y}_{\text{oss}}\right)$$



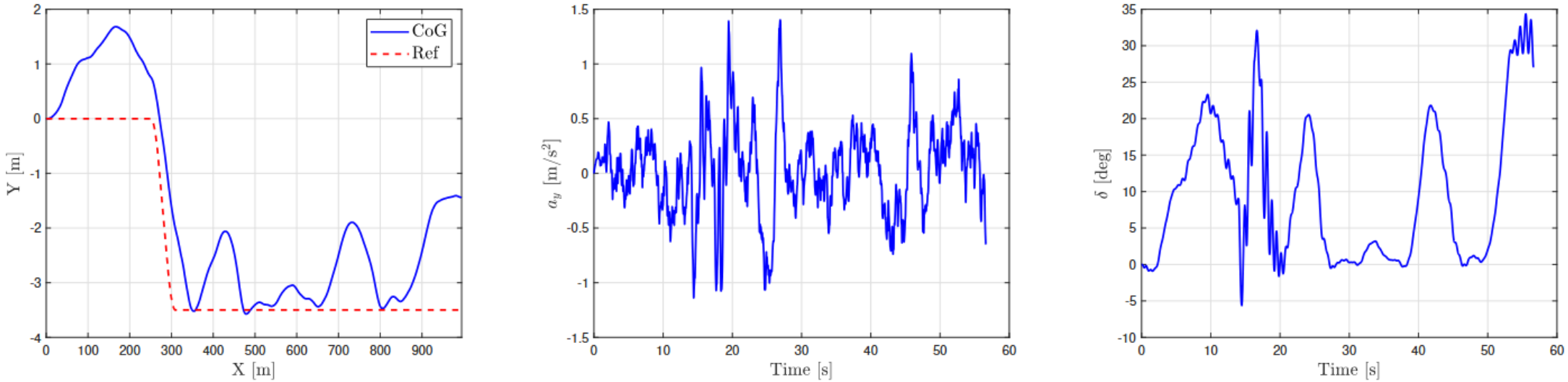
(a) Trajectory. (b) Lateral acceleration. (c) Steering angle.

Right turn - Dry road - Wind ($V = 30m/s$, $I_u = 14\%, L_u = 150m$)

4. Results

Influence of turning maneuvers on rollover speed

$$\delta = k_p \left(Y_{\text{ref}} - Y_{\text{oss}}\right) + k_d \left(\dot{Y}_{\text{ref}} - \dot{Y}_{\text{oss}}\right)$$

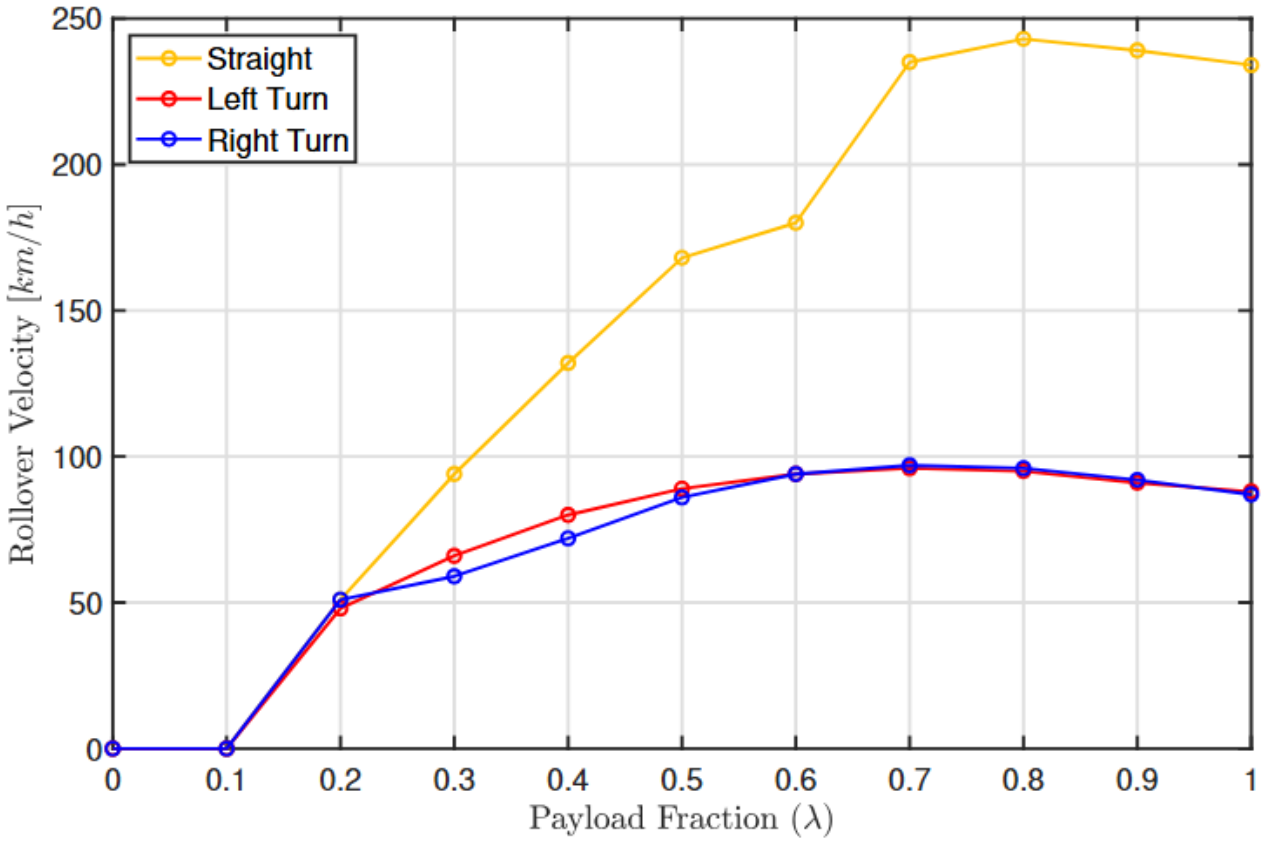


(a) Trajectory. (b) Lateral acceleration. (c) Steering angle.

Right turn - Snow road - Wind ($V = 30m/s$, $I_u = 14\%, L_u = 150m$)

4. Results

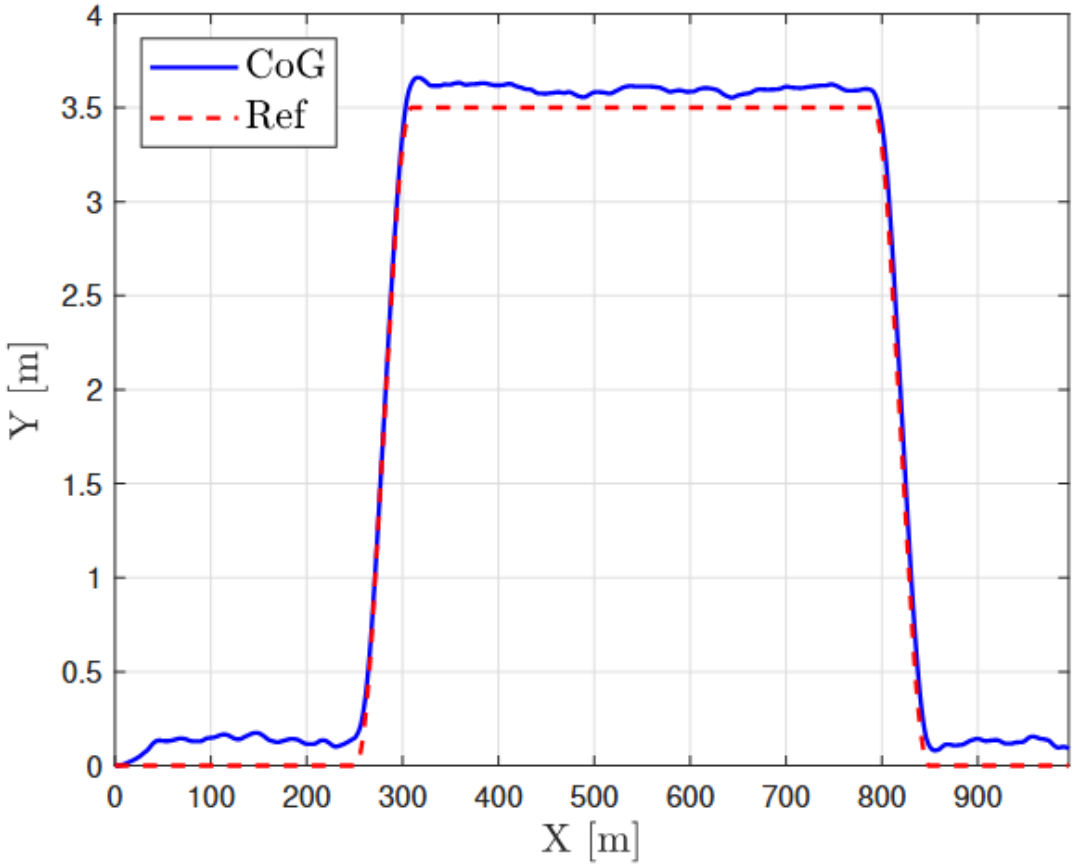
Influence of turning maneuvers on rollover speed



Rollover speed as function of payload in dry road condition and with a cross wind of 25 m/s and a turbulent index of 7%

4. Results

Influence of turning maneuvers on rollover speed



Trajectory of the truck following a double lane change (overtaking)

5. Conclusions

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CWC curves

- Effective method
- Safety thresholds

Road conditions

- Rollover delay with low friction condition
- Security with high friction condition

Payload distribution

- Instabilities with low λ

5. Conclusions

Roll stiffness distribution

- Importance in tuning suspension

Vehicle maneuvering

- Rollover speed decrease during turns

Multidimensional problem

- Road conditions, aerodynamic loading, internal vehicle dynamics…

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THANKS FOR YOUR ATTENTION